10th GRADE CHEMISTRY · TERM 1 · 1st COMMON WRITTEN EXAM

STUDY GUIDE — SENARYO 3

Your exam has 7 questions: ① Physical vs chemical change (KİM.10.1.1) → **1 Q** · ② Particle model of reactions (10.1.2) → **1 Q** · ③ Reaction types (10.1.3) → **2 Q** · ④ The mole concept (10.1.4) → **1 Q** · ⑤ Mole calculations (10.1.5) → **2 Q**.
3 of the 7 questions are about the **mole**. The paper is in Turkish, so the Turkish exam words are given in [brackets].
Atomic masses on the exam: H 1 · C 12 · N 14 · O 16 · Na 23 · S 32 · Cl 35.5 · $N_A = 6.02\times10^{23}$.

STATE SYMBOLS — TURKISH VS ENGLISH (EASY POINTS LOST HERE): solid = **(k)** katı · liquid = **(s)** sıvı · gas = **(g)** gaz · aqueous = **(suda)**. In Turkish, **(s) means LIQUID**, not solid. Water is $H_2O(s)$, salt crystals are NaCl(k).

1. PHYSICAL vs CHEMICAL CHANGE [fiziksel / kimyasal değişim] (1 Q)

	Physical change	Chemical change
What changes?	Only shape, size or state. The substance stays the same (same formula).	A **new substance** with new properties forms (new formula).
Bonds	Only forces *between* particles change [zayıf etkileşimler]	Bonds *inside* particles break and new ones form [güçlü etkileşimler]
Examples	Melting ice, boiling water, sublimation of naphthalene [naftalin], dissolving salt or sugar, breaking glass, cutting paper, mixing	Rusting, burning, cooking an egg, yogurt fermentation [mayalanma], milk souring, silver tarnishing, fireworks, electrolysis, photosynthesis, digestion

Evidence of a chemical change [kimyasal değişimin kanıtları]: color change · gas bubbles · a precipitate forms [çökelek / bulanıklık] · light, heat or sound is released (or heat is absorbed) · new smell or taste · usually hard to reverse.

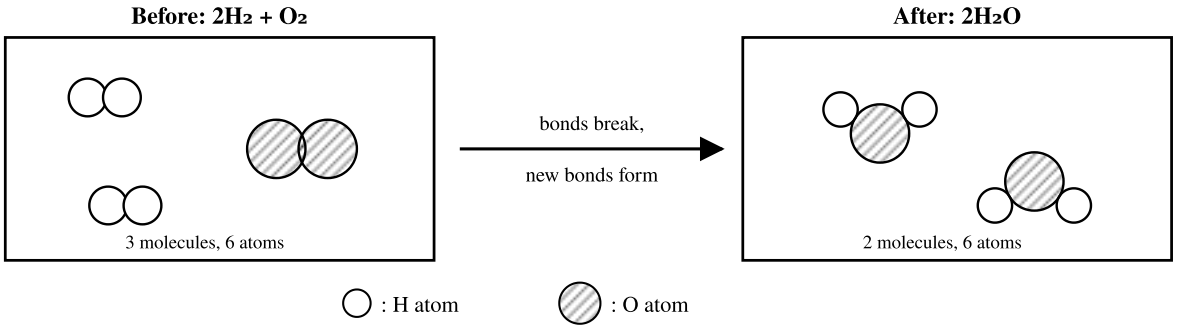
- **Traps:** energy change alone is **not** proof (melting also absorbs heat). Dissolving salt in water counts as **physical** in your course. Iodine turning into violet vapor is sublimation, which is physical.
- Breath into limewater [kireç suyu] → cloudy: $CO_2 + Ca(OH)_2 \rightarrow CaCO_3(k) + H_2O$. A precipitate means a **chemical** change.

Electrolysis of water [suyun elektrolizi]

$$\mathbf{2H_2O(s) \xrightarrow{\text{(electric current)}} 2H_2(g) + O_2(g)}$$

- **Evidence:** gas bubbles form at both electrodes. H_2 forms at the (–) electrode and O_2 at the (+) electrode, with a volume ratio of **H_2 : O_2 = 2 : 1**.
- **Tests:** H_2 burns with a "pop" sound when a lit splint is brought near it; O_2 re-ignites a glowing splint [kor hâlindeki kıymık].
- This is a chemical change because a compound is broken down into new substances (elements). The energy input is electrical.

2. PARTICLE MODEL OF REACTIONS [tepkimelerin tanecik modeli] (1 Q)



Stage	What happens to the particles	Energy
Before [önce]	Reactant particles (H_2 and O_2 molecules) move and **collide**. Only collisions with **enough energy** and the **right orientation** work [etkin çarpışma].	—
During [sırasında]	H–H and O=O **bonds break**; atoms separate and **rearrange**.	Breaking bonds **absorbs** energy
After [sonra]	New O–H **bonds form** → H_2O molecules (product).	Forming bonds **releases** energy

- **Always conserved [korunur]:** number and type of atoms, total mass, total charge.
- **Not necessarily conserved:** number of molecules (3 → 2 above), volume, physical state.
- If more energy is released by forming bonds than absorbed by breaking them, the reaction is **exothermic** [ekzotermik, ısı veren]; otherwise **endothermic** [endotermik, ısı alan].

3. REACTION TYPES [kimyasal tepkime türleri] (2 Q)

3.1 Combustion, synthesis, decomposition

- **Combustion [yanma]:** a substance reacts with O_2. Organic fuels give CO_2 + H_2O: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(s)$. Metals: $2Mg(k) + O_2(g) \rightarrow 2MgO(k)$. Combustion is exothermic (exception: N_2 combustion is endothermic).
- **Synthesis [sentez / oluşum]:** A + B → AB, e.g. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$.
- **Decomposition [analiz / ayrışma]:** AB → A + B, e.g. $CaCO_3(k) \rightarrow CaO(k) + CO_2(g)$; electrolysis of water.

3.2 Acid–base (neutralization) [asit-baz / nötralleşme]

acid + base → salt + water | $HCl(suda) + NaOH(suda) \rightarrow NaCl(suda) + H_2O(s)$

- **Net ionic equation [net iyon denklemi]:** $H^+(suda) + OH^-(suda) \rightarrow H_2O(s)$. The same for every strong acid + strong base.
- **Spectator ions [seyirci iyonlar]:** ions that appear unchanged on both sides (here Na^+ and Cl^-).
- **pH:** acid < 7, neutral = 7, base > 7 (25 °C). When a base neutralizes an acid, H^+ decreases and the **pH rises**. Neutralization is exothermic.
- Real life: antacid for heartburn, $Mg(OH)_2(k) + 2HCl(suda) \rightarrow MgCl_2(suda) + 2H_2O(s)$; adding lime to acidic soil.

3.3 Precipitation [çözünme-çökelme]

Two clear ionic solutions are mixed and an **insoluble solid** (precipitate [çökelek]) forms.

Full equation	Net ionic equation	Precipitate	Spectators
$Pb(NO_3)_2$(suda) + 2KI(suda) → PbI_2(k) + $2KNO_3$(suda)	$Pb^{2+} + 2I^- \rightarrow PbI_2$(k)	**yellow**	K^+, NO_3^-
$AgNO_3$(suda) + NaCl(suda) → AgCl(k) + $NaNO_3$(suda)	$Ag^+ + Cl^- \rightarrow$ AgCl(k)	white	Na^+, NO_3^-
$Ba(NO_3)_2$(suda) + Na_2SO_4(suda) → $BaSO_4$(k) + $2NaNO_3$(suda)	$Ba^{2+} + SO_4^{2-} \rightarrow$ $BaSO_4$(k)	white	Na^+, NO_3^-

- **Writing a net ionic equation in 4 steps:** ① write the balanced full equation with states → ② split every (suda) ionic compound into its ions → ③ cross out ions that are the same on both sides (spectators) → ④ write what remains.
- **Solubility rules (enough for this exam):** compounds of Na^+, K^+, NH_4^+ and NO_3^- are always soluble. Chlorides are soluble except AgCl and $PbCl_2$. Sulfates are soluble except $BaSO_4$ and $PbSO_4$. Most hydroxides and carbonates are insoluble (except Na^+, K^+, NH_4^+).

3.4 Redox [indirgenme-yükseltgenme]

A redox reaction is an **electron transfer**. **Oxidation [yükseltgenme]** = losing electrons, so the oxidation number goes **up**. **Reduction [indirgenme]** = gaining electrons, so the oxidation number goes **down**. Memory trick: **OIL RIG** (Oxidation Is Loss, Reduction Is Gain). Electrons lost = electrons gained.

> **OXIDATION NUMBER [YÜKSELTGENME BASAMAĞI] RULES:** element (Fe, O_2, Cu) = **0** · monatomic ion = its charge (Cu^{2+} = +2) · O is usually **−2** · H is usually **+1** · group 1 metals +1, group 2 metals +2 · the sum in a neutral compound = 0, in an ion = the ion's charge. Example: S in H_2SO_4 → 2(+1) + S + 4(−2) = 0 → S = **+6**.

Zn(k) + $CuSO_4$(suda) → $ZnSO_4$(suda) + Cu(k) → net: Zn(k) + Cu^{2+}(suda) → Zn^{2+}(suda) + Cu(k)

- Zn: 0 → +2, loses $2e^-$ → **oxidized** ($Zn \rightarrow Zn^{2+} + 2e^-$). Cu: +2 → 0, gains $2e^-$ → **reduced** ($Cu^{2+} + 2e^- \rightarrow$ Cu). SO_4^{2-} is a spectator.
- Metal + acid: Mg(k) + 2HCl(suda) → $MgCl_2$(suda) + H_2(g). Mg 0 → +2 (oxidized), H +1 → 0 (reduced). Gas bubbles are the evidence.
- **Which types are redox?** Combustion, metal + acid, metal + metal ion (single displacement), and synthesis/decomposition that involve elements → **redox**. Acid–base and precipitation → **not redox** (no oxidation number changes).

3.5 Summary — how to recognize the type

Type	Clue in the question	Net ionic / key form	Redox?
Combustion	"burns", reacts with O_2, flame, heat and light	fuel + O_2 → oxides	Yes
Acid–base	acid + base, "neutralizes", pH changes, antacid	$H^+ + OH^- \rightarrow H_2O$	No
Precipitation	two solutions mixed, "a white/yellow solid forms", cloudiness	cation + anion → solid(k)	No
Redox	metal into acid or salt solution, gas or metal deposit, electrons transferred	$M + X^{n+} \rightarrow M^{n+} + X$	Yes

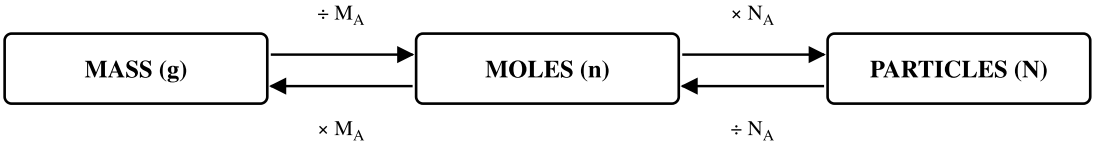
4. THE MOLE CONCEPT [mol kavramı] (1 Q)

- Atoms are far too small to count one by one, so chemists count them in packs. 1 dozen = 12, 1 pair = 2, 1 ream = 500 sheets, and **1 mole = 6.02×10^{23} particles** (Avogadro's number, N_A).
- **Rice analogy (asked on your mock):** if 100 grains weigh 2.5 g, one grain is 0.025 g and 1 mol of rice weighs $0.025 \times 6.02\times10^{23} \approx 1.5\times10^{22}$ g, which is absurdly large. So the mole is only useful for **tiny particles** (atoms,

molecules, ions). Grains are not identical, so the result is an **average**; atomic masses are averages for the same reason.

- **Always say "1 mol of WHAT":** 1 mol H_2O = 1 mol of molecules = 2 mol H atoms + 1 mol O atoms = **3 mol atoms**. 1 mol O_2 = 1 mol molecules = 2 mol O atoms.
- **Atomic mass unit [akb]:** 1 amu = 1/12 of the mass of one C-12 atom. 1 g = 6.02×10^{23} amu. One C atom = 12 amu; its mass in grams is 12 / 6.02×10^{23} ≈ 1.99×10^{-23} g.
- **Average atomic mass [ortalama atom kütlesi]:** weighted average of isotopes. Cl: 75% ^{35}Cl + 25% ^{37}Cl → 0.75×35 + 0.25×37 = **35.5**.
- **Molar mass [mol kütlesi, M_A]** = the mass of 1 mol, in g/mol. Add up the atomic masses: H_2O = 2(1) + 16 = **18** · CO_2 = 12 + 2(16) = **44** · NH_3 = **17** · CH_4 = **16** · O_2 = **32** · SO_2 = **64** · H_2SO_4 = **98** · NaCl = **58.5** · $C_6H_{12}O_6$ = **180**.

5. MOLE CALCULATIONS [mol hesaplamaları] (2 Q)



$n = m / M_A$ **$n = N / N_A$** **(atoms) = (molecules) × (atoms per molecule)**

Worked examples (new numbers, not the ones on your mock)

Question	Solution
1. 9 g H_2O: moles, molecules, atoms?	n = 9 / 18 = **0.5 mol** molecules → 0.5 × 6.02×10^{23} = **3.01×10^{23}** molecules. 3 atoms per molecule → 0.5 × 3 = **1.5 mol atoms** = 1.5 × 6.02×10^{23} = **9.03×10^{23} atoms**. H: 1 mol = 1 g; O: 0.5 mol = 8 g (1 + 8 = 9 ✓).
2. 22 g CO_2: moles? mass of O?	n = 22 / 44 = **0.5 mol**. O atoms = 0.5 × 2 = 1 mol → 1 × 16 = **16 g O**; C = 0.5 × 12 = 6 g (6 + 16 = 22 ✓).
3. 0.2 mol NH_3: mass, atoms, H atoms?	m = 0.2 × 17 = **3.4 g**. 4 atoms per molecule → 0.2 × 4 = **0.8 mol atoms**. H = 0.2 × 3 = 0.6 mol = 0.6 × 6.02×10^{23} = **3.612×10^{23} H atoms**.
4. 0.5 mol H_2SO_4: mass, mass of O?	m = 0.5 × 98 = **49 g**. O = 0.5 × 4 = 2 mol → **32 g**. Check: H 1 g + S 16 g + O 32 g = 49 g ✓.
5. 1.204×10^{24} CO_2 molecules: moles, mass?	n = 1.204×10^{24} / 6.02×10^{23} = **2 mol** → 2 × 44 = **88 g**. Atoms: 2 × 3 = 6 mol.
6. 8 g each of H_2, CH_4, O_2: which has the most molecules?	H_2: 8/2 = 4 mol · CH_4: 8/16 = 0.5 mol · O_2: 8/32 = 0.25 mol → **H_2 > CH_4 > O_2**. Equal mass does **not** mean equal particles; the smaller M_A, the more particles.
7. 0.25 mol of gas X weighs 11 g. M_A?	M_A = m / n = 11 / 0.25 = **44 g/mol** (it could be CO_2).
8. Mass of one H_2O molecule?	18 / 6.02×10^{23} ≈ **2.99×10^{-23} g**.

COMMON MISTAKES (EACH ONE COSTS POINTS): mixing up mol of molecules and mol of atoms · forgetting the subscript (H_2SO_4 has **4** O, $C_6H_{12}O_6$ has **24** atoms) · using 16 for O_2 instead of 32 · writing 12.04×10^{23} instead of **1.204×10^{24}** (correct scientific notation) · no units (always write g, mol, g/mol, "atoms") · not showing the formula: write $n = m / M_A$ first, then substitute.

ONLY IF YOUR TEACHER COVERED IT: at normal conditions [normal koşullar, NŞA] 1 mol of any gas = **22.4 L**, so n = V / 22.4. It is not part of your mock.

6. SENARYO 3 CHECKLIST — can you do these without looking?

☐ Sort 6–7 everyday changes into physical/chemical and give the evidence for each chemical one.
☐ Write the electrolysis of water with states and give its evidence.
☐ Draw a reaction with circles (before / after) and explain collide → break bonds (absorb) → form bonds (release).
☐ Say what is always conserved (atoms, mass, charge) and what is not (molecule count).
☐ Name the type of a reaction from a short story and write its balanced equation with (k)(s)(g)(suda).
☐ Write a net ionic equation and name the spectator ions.
☐ Find oxidation numbers and say which species is oxidized and which is reduced.
☐ Explain why the mole is used (rice analogy) and the difference between mol atoms and mol molecules.
☐ Convert mass ⇄ moles ⇄ particles, and count atoms inside a molecule.

Sources: your chemistry worksheets and mole questions, Aydın Miray Senaryo 3 papers, MEB "Yazılıya Hazırlanıyorum" Kimya 10 style. Not an official document.